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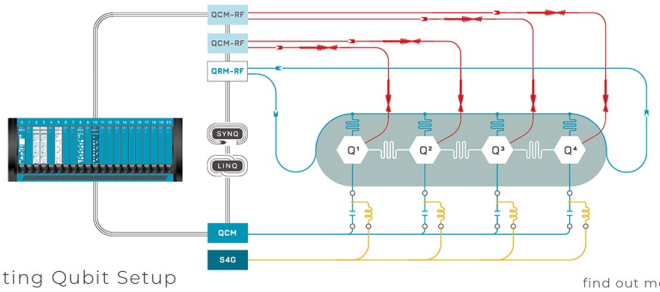
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ABSTRACT

Manipulating heat fluxes at the nanoscale has become increasingly important in modern microelectronics. However, many methods of heat manipulations rely on complex nanofabrication. Here, we propose simple designs for collimation and focusing of thermal phonons based on parabolic mirrors that require no nanofabrication. We perform Monte Carlo simulations of a ballistic phonon transport in silicon membranes with parabolic boundaries. Our simulations demonstrate that parabolic surfaces can act as parabolic mirrors for phonons, thus collimating or focusing phonon fluxes in semiconductors. Such parabolic mirrors can create a directional flux of thermal phonons emitted from a nanoscale hot spot or focus a collimated phonon flux into a hot spot. These devices open new possibilities in the thermal management of low-temperature systems, such as quantum circuits or cryogenic particle detectors.

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Manipulating heat fluxes in semiconductors becomes increasingly important as the miniaturization of modern electronics approaches atomic limits.[1] Over the past decade, researchers explored various heat manipulation methods, including wave phononics, which promised control over heat fluxes via interference of phonons carrying heat.[2–4] Indeed, like photons in photonic crystals, phonons can be guided in specific directions or focused into a spot using structures with periodic patterning called phononic crystals.[5–7] However, wave phononics appears to be efficient only at sub-kelvin temperatures[8–10] or in atomic-scale superlattices,[11–13] where a substantial portion of phonons can experience elastic surface scattering.

Recently, we proposed an alternative approach called ray phononics,[14] which uses ballistic phonon transport instead of phonon wave interference. Ray phononics suggests that certain nanopatterns tend to scatter phonons in a certain direction, which enables phonon guiding,[14] collimation,[15] shielding,[16] and focusing.[15,17] Yet, the ray phononics approach still relies on the complex fabrication of nanopatterns using expensive lithography techniques.[14] In this work, we propose simplified collimation and focusing devices that require less nanopatterning and can be partly fabricated with simpler lithography techniques.

Our devices rely on ballistic phonon transport and specular reflections from a parabolic surface. Parabolic surfaces collect the parallel beams into the focal point or likewise make the beams emitted from the focal point parallel, as shown in Fig. 1(a). Such parabolic mirrors are widely used in ray optics for collimation and focusing of light. However, the wavelength of phonons that carry heat is two orders shorter than the wavelength of photons that carry light. Thus, it remains unclear whether thermal phonons can experience enough of specular reflections to produce similar effects in realistic structures. Moreover, the effects are only expected if the internal phonon scattering is weak. Thus, temperature dependence of these effects also remains unclear. Here, we aim to demonstrate that under certain conditions, parabolic lenses can, indeed, collimate phonons emitted from a hot spot and focus a collimated phonon flux into a spot.

To demonstrate heat flux manipulation using parabolic mirrors, we use the Monte Carlo method.[18–20] Our algorithm simulates the ballistic transport of 40 000 phonons in a three-dimensional model of a 150-nm-thick silicon membrane with a parabolic surface, illustrated in Fig. 1(b). The algorithm traces the phonon trajectories through the system [Fig. 1(c)] and records the phonon position at every time step, which is used to construct the phonon energy map [Fig. 1(d)]. The model generates phonon at a given temperature according to the corresponding Plank distribution and assigns phonon properties according to the bulk phonon dispersion relation in silicon at a given frequency. All phonons are launched from either a hot spot (in the case of the collimation) or the hot side (in the case of the focusing) in a random direction. Inside the system, the phonon can undergo surface scattering and internal phonon scattering. The model accounts for internal phonon scattering using the Matthiessen rule. In the Matthiessen rule, the impurity scattering is given by $\tau_I^{-1} = A\omega^4$, with

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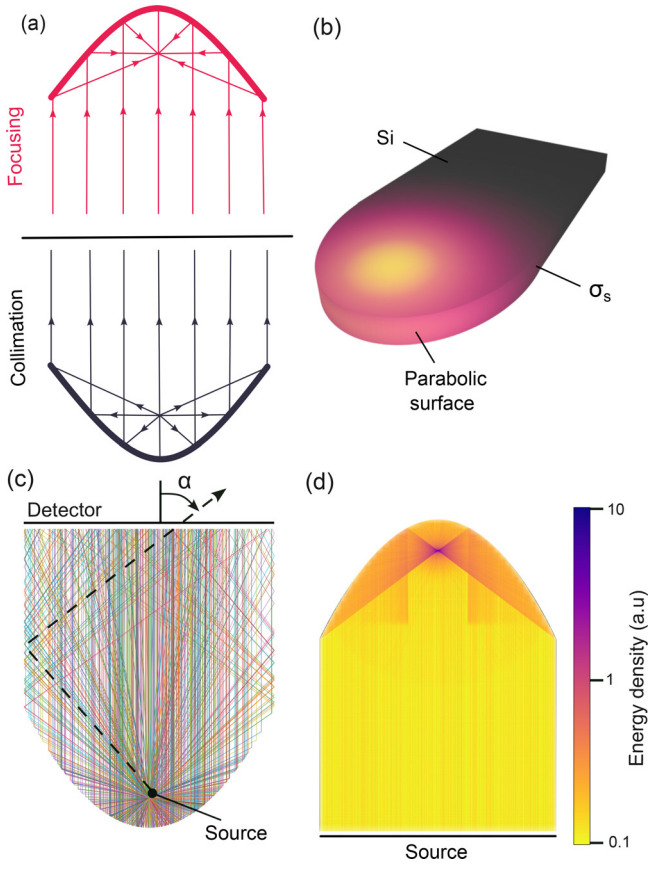


FIG. 1. (a) Schemes of collimation and focusing using a parabolic mirror. (b) Three-dimensional model of the structure and (c) examples of phonon trajectories showing collimation. (d) Example of the energy map depicting focusing effect in the ideal case of a collimated phonon flux.

$A = 2.95 \times 10^{-45}$ and Umpklapp scattering is given by $\tau_U^{-1} = B\omega^2 T \exp(-T_d/T)$, with $B = 0.95 \times 10^{-19}$ and $T_d = 152$ K.

The surface scattering can be either diffuse or specular depending on the phonon wavelength λ, angle of incidence ϕ, and the surface roughness σ. In this work, the surface roughness of the top and bottom surfaces is set to 0.2 nm,[21,22] while the roughness of the sidewalls is set to 2 nm[10] unless otherwise noted. The probability of phonon to undergo specular scattering is given by[23–25]

$$p = \exp(-16\pi^2\sigma^2 \cos^2\phi/\lambda^2). \quad (1)$$

Once the phonons reach the cold side, their angles to the y axis are recorded to construct the distribution of exit angles. More details on the Monte Carlo algorithm are available in our previous works[14,20] and in the code repository.

We begin by demonstrating the collimation effect and its dependence on the sidewall roughness σ_s, source size, and temperature. The intensity of the collimation effect is quantified using the exit angle distribution of phonons as they are detected by the detector at the end of the system. This distribution represents the directions of phonon flights when they leave the system and is expected to peak at 0° if most phonons are indeed collimated by the parabolic surface.

Figure 2(a) shows the exit-angle distribution of phonons as a function of sidewall roughness at 4 K. The distributions peak at 0° meaning that the parabolic surface indeed collimates thermal phonons into a vertical flux. Naturally, the effect is strongest for the case of

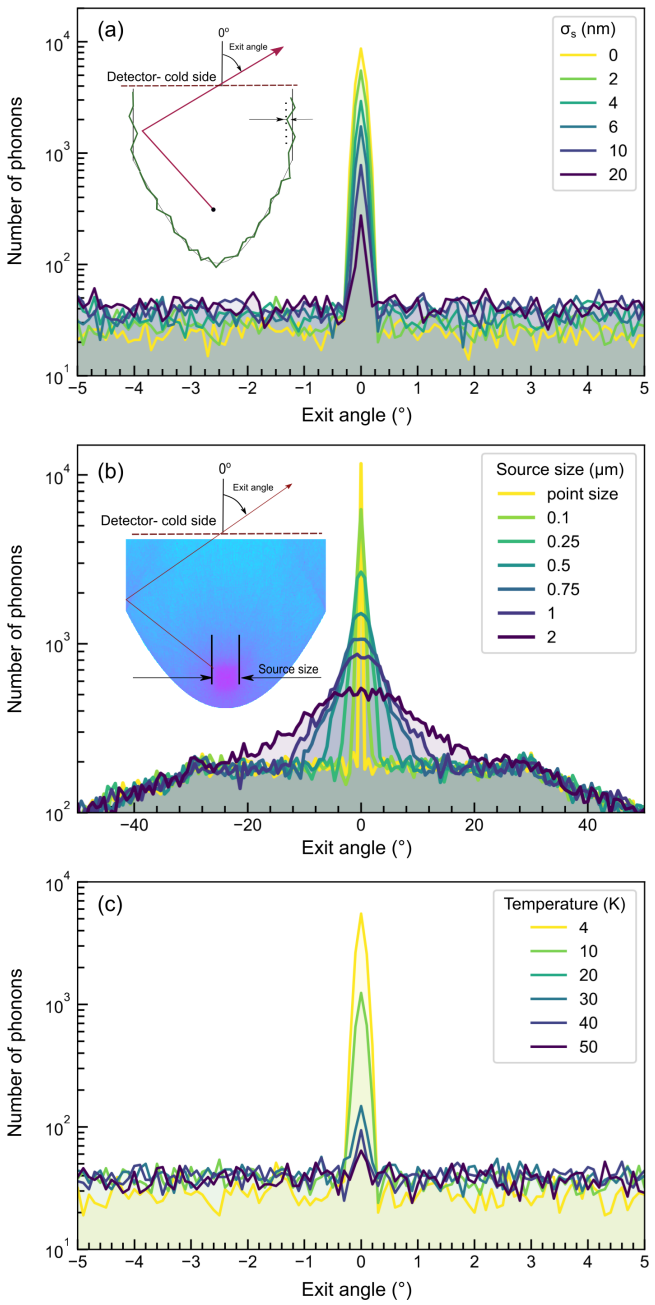


FIG. 2. Exit angle distribution for different (a) sidewall surface roughnesses, (b) source size, and (c) temperature of the system. The source is located at the focus of the parabolic surface. The peak at 0° is indicative of a collimation effect. For (a) and (b), the temperature of the system is 4 K. For (a) and (c), the source is point size.

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perfectly specular reflections ($\sigma_s = 0$ nm), albeit not entirely ideal as some phonons are never reflected from the parabola. As roughness is increased, the peak weakens, decreasing by more than 90% for the roughness of 10 nm.

The key behind the collimation effect is dominant specular scattering from the parabolic boundary, as it gives mirror-like reflection in the desired direction. For rougher surfaces, the probability of specular scattering decreases while the probability of diffuse scattering increases—randomizing the post-scattering direction of phonons. At $\sigma_s = 0$ nm, 100% of phonons undergo specular scattering, at $\sigma_s = 4$ nm, it reduces to 42%, and at $\sigma_s = 10$ nm, the scattering is mostly diffuse (77%). These simulations show that surfaces with realistic roughness or 0.2–3 nm can produce the collimated flux.

Next, we consider the size of the hot spot and study how the collimation effect is affected by a non-ideal source of phonons at 4 K. Figure 2(b) compares the collimation effect for the phonon sources of different sizes. Naturally, the peak is highest for the point source. For the 100 nm source, the peak is still prominent but smaller than that of the point source. As the source size is increased, the effect weakens and almost vanishes for the source of 2 μm in size. Since phonons must initiate from the focus of the parabola to be collimated precisely parallel to the vertical axis, displacement of initial position from the focus proportionally weakens the collimation effect. However, a realistic point source of a few hundred nanometers in size can produce the collimation effect.

Finally, we study the impact of the temperature on the collimation effect. Increase in temperature shifts the phonon spectrum toward higher frequencies. Thus, since the specularity is inversely proportional to phonon frequency [Eq. (1)], the probability of specular boundary scattering decreases with temperature. Moreover, an increase in temperature also causes an increase in the frequency of internal scattering events.

At 4 K, the specular scattering is dominant, and therefore, reflection from the parabolic surface results in the collimation effect, as revealed by a strong peak in Fig. 2(c). At 10 K, internal scattering remains negligible, but diffuse boundary scattering has increased, causing the weakening of the peak. As temperature is increased further, the diffuse boundary scattering continues to increase, whereas internal scattering also starts playing a role. At 50 K, about 43% of all scattering events are caused by the internal scattering and 50% by diffuse boundary scattering. In other words, specular scattering of phonons becomes unlikely at 50 K, resulting in the flat distribution of exit angles. Thus, the collimation effect in silicon seems to be limited to low temperatures.

Let us now consider an inverse effect—the focusing of the phonon flux with the parabolic surface. For the focusing effect to occur, the incoming flux needs to be collimated parallel to the axis of the parabola. One method to create such a flux is to use the collimation effect established above. Figure 3(a) shows the energy map of a two-sided parabolic mirror structure. The phonons launched from the source become collimated after scattering from the bottom parabolic surface. These directional phonons then undergo scattering from the top parabolic surface, which results in the concentration of phonons at the focal point. Figure 3(a) (left) shows that the energy density along the vertical axis displays two peaks. The primary peak near the bottom corresponds to the source, whereas the second peak is due to the focusing of thermal phonons at the focal point.

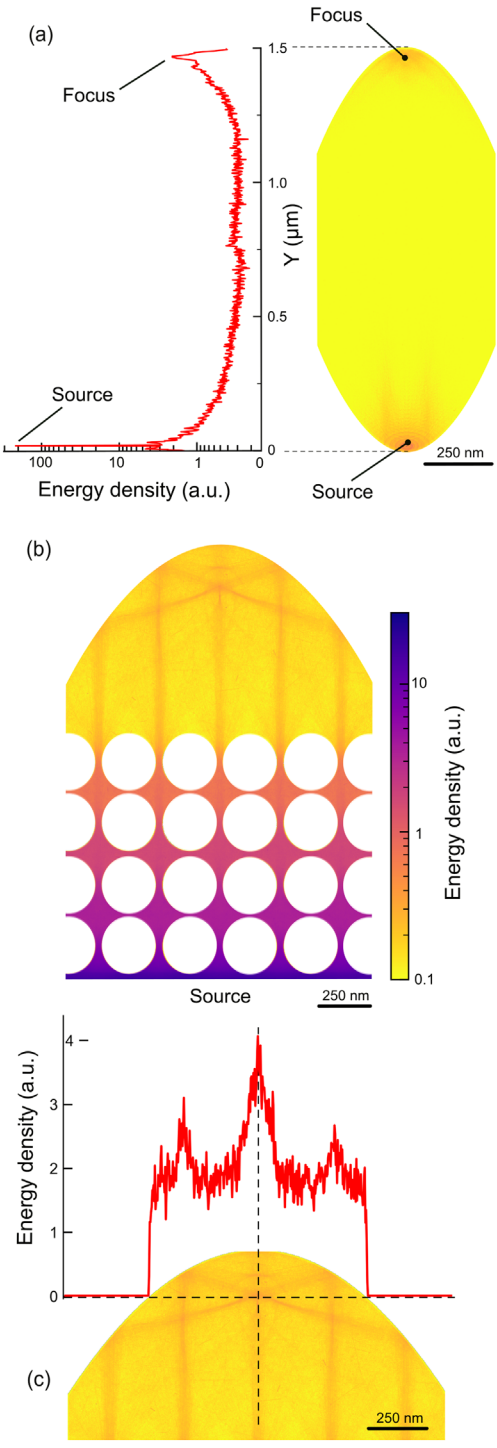


FIG. 3. Focusing using parabolic thermal mirrors at 4 K. (a) Energy density map of a structure with parabolic surfaces as a source and a slice of the energy density along the central axis of the structure. The first peak is caused by the source and the second peak is caused by the focusing of phonons. (b) Energy density map of a structure with a phononic crystal as a source. (c) Energy density at the focal point of the parabola.

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Another method to create a collimated flux is by using an array of holes. Due to the presence of holes with narrow passages (30 nm) between them, phonons are forced to become directed along the passage between those holes, as has been demonstrated previously[14,15,19] As a result, the phonons are emitted from the array of holes in the direction parallel to the passages between the holes, as shown in Fig. 3(b). The emitted phonon fluxes are then focused by the parabolic surface into a hot spot. Figure 3(c) shows the close-up view of the focusing effect and the energy density around the hot spot. The peak of energy density at the hot spot is about twice higher than the surrounding, which represents the focusing effect.

In conclusion, we demonstrated how parabolic surfaces could be used to collimate and focus fluxes of thermal phonons in solids. Our simulations showed that heat emitted from a nanoscale hot spot could be collimated into a heat flux oriented in a desired direction. Moreover, the simulations demonstrated how a parabolic surface could focus a collimated heat flux back into a hot spot. The achieved focusing effect is weaker than the focusing achieved using a thermal lens structure made of arrays of holes.[15,17] Yet, the efficiency of such parabolic mirrors increases for smoother surfaces, smaller hot spots, and lower temperatures.

An experimental demonstration is expected to be possible with a hot spot of a few hundred nanometers in silicon membranes with a surface roughness below 2 nm at the temperature of a few kelvins. The experiments might be complicated by the fact that thermal phonons do not contribute to local temperature along their ballistic path and only thermalize at the point of diffuse scattering. Thus, a design of experimental samples may include a scattering point, such as a hole or a defect, at the focal point of the focusing lens. Materials with longer phonon mean free paths, like silicon carbide[26] or boron arsenide,[27] may improve the efficiency of parabolic mirrors. Thus, future works should explore alternative materials and configurations to enable ray phononic heat flux manipulations at higher temperatures.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Dhanishtha Singh: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Software (equal); Visualization (equal); Writing – original draft (equal). **Roman Anufriev:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Software (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Masahiro Nomura:** Conceptualization (equal); Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data are available from the authors upon request. The Monte Carlo code is available at https://github.com/anufrievroman/freepaths.

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